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Since the leftmost bit is a carry, we add 1 to the result, giving us 1010. However, we discard the leftmost carry bit, resulting in 010.

275. The correct answer is a) To subtract  $8_{10}$  from  $4_{10}$  using 1's complement, we first convert the numbers to their 1's complement representation.

$4_{10}$  is represented as 0100 in 1's complement.

$8_{10}$  is represented as 1000 in 1's complement.

Next, we perform the subtraction using regular addition.  $0100 + 1000 = 1100$ .

Since the leftmost bit is a carry, we add 1 to the result, giving us 1101. However, we discard the leftmost carry bit, resulting in 1000.

276. The correct answer is b) 0110

To subtract  $1_{10}$  from  $7_{10}$  using 2's complement, we first represent the numbers in 2's complement form.

$7_{10}$  is represented as 0111 in 4-bit 2's complement.

$1_{10}$  is represented as 0001 in 4-bit 2's complement.

Next, we perform the subtraction by adding the 2's complement of the subtrahend to the minuend.

$0111 + 1111 = 10110$  (Discard the carry bit)

277. The correct answer is a) To subtract  $5_{10}$  from  $7_{10}$  using 2's complement, we first represent the numbers in 2's complement form.

$7_{10}$  is represented as 0111 in 4-bit 2's complement.

$5_{10}$  is represented as 0101 in 4-bit 2's complement.

Next, we perform the subtraction by adding the 2's complement of the subtrahend to the minuend.

$0111 + 1011 = 10010$  (Discard the carry bit)

278. Answer: The correct answer is d) To subtract  $2_{10}$  from  $7_{10}$  using 2's complement, we first represent the numbers in 2's complement form.

$7_{10}$  is represented as 0111 in 4-bit 2's complement.

$2_{10}$  is represented as 0010 in 4-bit 2's complement.

Next, we perform the subtraction by adding the 2's complement of the subtrahend to the minuend.

$0111 + 1110 = 10101$  (Discard the carry bit)

279. The correct answer is d)

280. The correct answer is a)

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281. The correct answer is a) In Gray code, two successive values differ by only one bit, making it useful for minimizing errors during binary transitions.

282. The correct answer is d) Gray code is a self-complementary numbering system, meaning that the complement of a Gray code number is also its corresponding code.

283. The correct answer is b) To represent 8 distinct values in Gray code, 3 bits are required. Each bit represents a different value in the Gray code sequence.

284. The correct answer is b) The Gray code representation of the binary number 1011 is 1110. In Gray code, each bit position changes only one bit at a time.

285. The correct answer is d) Gray code is commonly used for encoding rotary encoders, which are used in position sensing applications such as robotics and automation.

286. The correct answer is c) The complementarity property of Gray code ensures that only one bit changes at a time, reducing errors and glitches in digital systems.

287. The correct answer is b) A 4-bit Gray code can represent 8 different values. Each bit in the 4-bit sequence represents a unique value in the Gray code sequence.

288. The correct answer is c) Reflected Gray code is obtained by toggling only one bit at a time, ensuring that adjacent codes differ by only one bit transition.

289. The correct answer is b) One advantage of using Gray code in digital systems is its compact representation of numbers, requiring fewer transitions compared to binary code.

290. The correct answer is c) Gray code is commonly used in digital-to-analog converters to prevent glitches during transitions, ensuring a smooth and accurate analog output.



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5

## Boolean Algebra and Logic Gates

- What is the output of an AND gate if both inputs are 1?
  - 1
  - 0
  - X (undefined)
  - None of the above
- How many inputs does an AND gate typically have?
  - 1
  - 2
  - 3
  - It can vary
- What is the output of an AND gate if one input is 0 and the other input is 1?
  - 1
  - 0
  - X (undefined)
  - None of the above
- Which Boolean expression represents the operation of an AND gate?
  - $A + B$
  - $A * B$
  - $A \text{ XOR } B$
  - $A' + B'$
- What is the output of an AND gate if both inputs are 1?
  - 1
  - 0
  - X (undefined)
  - None of the above
- In a truth table, what are the possible combinations of inputs for an AND gate?
  - 0 and 1
  - 0 and 0
  - 1 and 1
  - All of the above
- What is the output of an AND gate if one or both inputs are X (undefined)?
  - 1
  - 0
  - X
  - None of the above
- In a logic circuit diagram, how is an AND gate typically represented?
  - Triangle
  - Rectangle
  - Circle
  - Diamond
- What is the result of an AND operation between any input and 0?
  - 0

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- b) 1  
c) X (undefined)  
d) None of the above

10. What is the truth table representation of an AND gate with two inputs?

a) A | B | Output

0 | 0 | 0

0 | 1 | 1

1 | 0 | 1

1 | 1 | 0

b) A | B | Output

0 | 0 | 0

0 | 1 | 0

1 | 0 | 0

1 | 1 | 1

c) A | B | Output

0 | 0 | 0

0 | 1 | 1

1 | 0 | 0

1 | 1 | 1

d) A | B | Output

0 | 0 | 1

0 | 1 | 1

1 | 0 | 1

1 | 1 | 0

11. What is the output of an OR gate if both inputs are 0?

- a) 0  
b) 1  
c) X (undefined)  
d) None of the above

12. How many inputs does an OR gate typically have?

- a) 1  
b) 2  
c) 3  
d) It can vary

13. What is the output of an OR gate if one input is 1 and the other input is 0?

- a) 0  
b) 1  
c) X (undefined)  
d) None of the above

14. Which Boolean expression represents the operation of an OR gate?

- a)  $A + B$   
b)  $A * B$   
c)  $A \text{ XOR } B$   
d)  $A' + B$

15. What is the output of an OR gate if both inputs are 1?

- a) 0  
b) 1  
c) X (undefined)  
d) None of the above

16. In a truth table, what are the possible combinations of inputs for an OR gate?

- a) 0 and 1  
b) 0 and 0  
c) 1 and 1

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d) All of the above

1 | 0 | 1  
1 | 1 | 0

17. What is the output of an OR gate if one or both inputs are X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

c) A | B | Output

0 | 0 | 0  
0 | 1 | 1  
1 | 0 | 1  
1 | 1 | 1

18. In a logic circuit diagram, how is an OR gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

d) A | B | Output

0 | 0 | 1  
0 | 1 | 0  
1 | 0 | 0  
1 | 1 | 0

Certainly! Here are some multiple-choice questions on NOT gates:

19. What is the result of an OR operation between any input and 1?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

21. What is the output of a NOT gate if the input is 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

20. What is the truth table representation of an OR gate with two inputs?

a) A | B | Output

0 | 0 | 0  
0 | 1 | 0  
1 | 0 | 0  
1 | 1 | 1

22. How many inputs does a NOT gate typically have?

- a) 1
- b) 2
- c) 3
- d) It can vary

b) A | B | Output

0 | 0 | 0  
0 | 1 | 1

23. What is the output of a NOT gate if the input is 1?

- a) 0
- b) 1
- c) X (undefined)

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d) None of the above

24. Which Boolean expression represents the operation of a NOT gate?

- a)  $A + B$
- b)  $A * B$
- c)  $A \text{ XOR } B$
- d)  $A'$

25. What is the output of a NOT gate if the input is X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

26. In a logic circuit diagram, how is a NOT gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

27. What is the result of applying a NOT operation twice ( $\text{NOT}(\text{NOT})$ ) to an input?

- a) The original input
- b) 0
- c) 1
- d) X (undefined)

28. In a truth table, what are the possible combinations of inputs for a NOT gate?

- a) 0

b) 1

- c) X (undefined)
- d) Both a) and b)

29. What is the output of a NOT gate if the input is X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

30. What is the truth table representation of a NOT gate?

| a) Input   Output |   |
|-------------------|---|
| 0                 | 0 |
| 1                 | 1 |

b) Input | Output

|   |   |
|---|---|
| 0 | 1 |
| 1 | 0 |

c) Input | Output

|   |   |
|---|---|
| 0 | 0 |
| 1 | 0 |

d) Input | Output

|   |   |
|---|---|
| 0 | 1 |
| 1 | 1 |

Remember, a NOT gate, also known as an inverter, produces the logical complement of its input. It produces an output of 1 if the input is 0 and vice versa.

31. What is the output of a NAND gate if both inputs are 0?

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- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

32. How many inputs does a NAND gate typically have?

- a) 1
- b) 2
- c) 3
- d) It can vary

33. What is the output of a NAND gate if one input is 1 and the other input is 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

34. Which Boolean expression represents the operation of a NAND gate?

- a)  $A + B$
- b)  $A * B$
- c)  $A \text{ XOR } B$
- d)  $(A * B)'$

35. What is the output of a NAND gate if both inputs are 1?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

36. In a truth table, what are the possible combinations of inputs for a NAND gate?

- a) 0 and 1
- b) 0 and 0
- c) 1 and 1
- d) All of the above

37. What is the output of a NAND gate if one or both inputs are X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

38. In a logic circuit diagram, how is a NAND gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

39. What is the result of applying a NAND operation twice (NAND(NAND)) to an input?

- a) The original input
- b) 0
- c) 1
- d) X (undefined)

40. What is the truth table representation of a NAND gate?

- a) A | B | Output
- 0 | 0 | 0
- 0 | 1 | 1

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1 | 0 | 1  
1 | 1 | 1

b) A | B | Output

0 | 0 | 1  
0 | 1 | 0  
1 | 0 | 0  
1 | 1 | 0

c) A | B | Output

0 | 0 | 1  
0 | 1 | 1  
1 | 0 | 1  
1 | 1 | 0

d) A | B | Output

0 | 0 | 0  
0 | 1 | 0  
1 | 0 | 0  
1 | 1 | 1

41. What is the output of a NOR gate if both inputs are 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

42. How many inputs does a NOR gate typically have?

- a) 1
- b) 2
- c) 3
- d) It can vary

43. What is the output of a NOR gate if one input is 1 and the other input is 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

44. Which Boolean expression represents the operation of a NOR gate?

- a)  $A + B$
- b)  $A * B$
- c)  $A \text{ XOR } B$
- d)  $(A + B)'$

45. What is the output of a NOR gate if both inputs are 1?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

46. In a truth table, what are the possible combinations of inputs for a NOR gate?

- a) 0 and 1
- b) 0 and 0
- c) 1 and 1
- d) All of the above

47. What is the output of a NOR gate if one or both inputs are X (undefined)?

- a) 0
- b) 1

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c) X

d) None of the above

48. In a logic circuit diagram, how is a NOR gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

49. What is the result of applying a NOR operation twice (NOR(NOR)) to an input?

- a) The original input
- b) 0
- c) 1
- d) X (undefined)

50. What is the truth table representation of a NOR gate?

a) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

b) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

c) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

d) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

51. What is the output of an XOR gate if both inputs are 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

52. How many inputs does an XOR gate typically have?

- a) 1
- b) 2
- c) 3
- d) It can vary

53. What is the output of an XOR gate if one input is 1 and the other input is 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

54. Which Boolean expression represents the operation of an XOR gate?

- a)  $A + B$
- b)  $A * B$
- c)  $A \text{ XOR } B$
- d)  $(A + B)'$

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55. What is the output of an XOR gate if both inputs are 1?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

56. In a truth table, what are the possible combinations of inputs for an XOR gate?

- a) 0 and 1
- b) 0 and 0
- c) 1 and 1
- d) All of the above

57. What is the output of an XOR gate if one or both inputs are X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

58. In a logic circuit diagram, how is an XOR gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

59. What is the result of applying an XOR operation twice (XOR(XOR)) to an input?

- a) The original input
- b) 0
- c) 1

d) X (undefined)

60. What is the truth table representation of an XOR gate?

a) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

b) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

c) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

d) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

61. What is the output of an XNOR gate if both inputs are 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

62. How many inputs does an XNOR gate typically have?

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- a) 1
- b) 2
- c) 3
- d) It can vary

63. What is the output of an XNOR gate if one input is 1 and the other input is 0?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

64. Which Boolean expression represents the operation of an XNOR gate?

- a)  $A + B$
- b)  $A * B$
- c)  $A \text{ XNOR } B$
- d)  $(A + B)'$

65. What is the output of an XNOR gate if both inputs are 1?

- a) 0
- b) 1
- c) X (undefined)
- d) None of the above

66. In a truth table, what are the possible combinations of inputs for an XNOR gate?

- a) 0 and 1
- b) 0 and 0
- c) 1 and 1
- d) All of the above

67. What is the output of an XNOR gate if one or both inputs are X (undefined)?

- a) 0
- b) 1
- c) X
- d) None of the above

68. In a logic circuit diagram, how is an XNOR gate typically represented?

- a) Triangle
- b) Rectangle
- c) Circle
- d) Diamond

69. What is the result of applying an XNOR operation twice ( $\text{XNOR}(\text{XNOR})$ ) to an input?

- a) The original input
- b) 0
- c) 1
- d) X (undefined)

70. What is the truth table representation of an XNOR gate?

a) A | B | Output

0 | 0 | 0

0 | 1 | 1

1 | 0 | 1

1 | 1 | 0

b) A | B | Output

0 | 0 | 1

0 | 1 | 0

1 | 0 | 0

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1 | 1 | 1

c) A | B | Output

0 | 0 | 0

0 | 1 | 0

1 | 0 | 0

1 | 1 | 1

d) A | B | Output

0 | 0 | 1

0 | 1 | 1

1 | 0 | 1

1 | 1 | 0

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## Explanation

1. The correct answer is a) 1.

An AND gate produces an output of 1 only when both of its inputs are 1. In this case, since both inputs are 1, the output of the AND gate will be 1.

2. The correct answer is b) 2. An AND gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

3. The correct answer is b) 0. An AND gate produces an output of 0 if any of its inputs is 0. In this case, one input is 0 while the other is 1, so the output of the AND gate will be 0.

4. The correct answer is b)  $A * B$ . The Boolean expression  $A * B$  represents the operation of an AND gate, where  $*$  denotes the logical AND operation.

5. The correct answer is b) 0. An AND gate produces an output of 0 if any of its inputs is 0. In this case, both inputs are 0, so the output of the AND gate will be 0.

6. The correct answer is d) All of the above. In a truth table for an AND gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 1.

7. The correct answer is b) 0. An AND gate produces an output of 0 if any of its inputs is 0. If one or both inputs are X (undefined), the output will be 0.

8. The correct answer is b) Rectangle. In logic circuit diagrams, an AND gate is typically represented by a rectangle shape with two input lines and one output line.

9. The correct answer is a) 0. An AND operation with any input and 0 will always result in 0. It follows the rule that anything ANDed with 0 is always 0.

10. The correct answer is c)  $A | B$  | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

This truth table represents an AND gate with two inputs, where the output is 1 only when both inputs

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are 1. In all other cases, the output is 0.

Remember, an AND gate performs the logical AND operation on its inputs and produces an output based on the defined behavior of the AND gate.

11. The correct answer is b) 1. An OR gate produces an output of 1 if at least one of its inputs is 1. In this case, both inputs are 0, so the output of the OR gate will be 0.

12. The correct answer is b) 2. An OR gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

13. The correct answer is b) 1. An OR gate produces an output of 1 if at least one of its inputs is 1. In this case, one input is 1 while the other is 0, so the output of the OR gate will be 1.

14. The correct answer is a)  $A + B$ . The Boolean expression  $A + B$  represents the operation of an OR gate, where  $+$  denotes the logical OR operation.

15. The correct answer is b) 1. An OR gate produces an output of 1 if at least one of its inputs is 1. In this

case, both inputs are 1, so the output of the OR gate will be 1.

16. The correct answer is d) All of the above. In a truth table for an OR gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 1.

17. The correct answer is c) X. If one or both inputs of an OR gate are X (undefined), the output will be X as well. This is because the output of an OR gate is determined by the inputs, and if the inputs are undefined, the output cannot be determined.

18. The correct answer is b) Rectangle. In logic circuit diagrams, an OR gate is typically represented by a rectangle shape with two input lines and one output line.

19. The correct answer is b) 1. An OR operation with any input and 1 will always result in 1. It follows the rule that anything ORed with 1 is always 1.

20. The correct answer is c)  $A \mid B \mid$   
Output

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

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This truth table represents an OR gate with two inputs, where the output is 1 if at least one input is 1. In this case, if either input A or input B is 1 or if both inputs are 1, the output will be 1.

Remember, an OR gate performs the logical OR operation on its inputs and produces an output based on the defined behavior of the OR gate.

21. The correct answer is b) 1. A NOT gate, also known as an inverter, produces the logical complement of its input. If the input is 0, the output of the NOT gate will be 1.

22. The correct answer is a) 1. A NOT gate typically has only one input. It takes a single signal or binary value as input and produces a single output.

23. The correct answer is a) 0. A NOT gate produces the logical complement of its input. If the input is 1, the output of the NOT gate will be 0.

24. The correct answer is d)  $A'$ . The Boolean expression  $A'$  represents the operation of a NOT gate. It denotes the logical complement or inversion of the input value.

25. The correct answer is c) X. If the input of a NOT gate is X (undefined), the output will also be X. This is because the output of a NOT gate is determined by the input value, and if the input is undefined, the output cannot be determined.

26. The correct answer is a) Triangle. In logic circuit diagrams, a NOT gate is typically represented by a triangle shape. The input is connected to the base of the triangle, and the output is taken from the tip of the triangle.

27. The correct answer is a) The original input. Applying the NOT operation twice ( $\text{NOT}(\text{NOT})$ ) to an input will result in the original input value. The NOT gate inverts the input, so inverting it again restores the original value.

28. The correct answer is d) Both a) and b). In a truth table for a NOT gate, there are two possible combinations of inputs: 0 and 1. The output will be the logical complement of the input value for both cases.

29. The correct answer is b) 1. If the input of a NOT gate is X (undefined), the output will also be X. This is

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because the output of a NOT gate is determined by the input value, and if the input is undefined, the output cannot be determined.

30. The correct answer is b) Input | Output

|   |  |   |
|---|--|---|
| 0 |  | 1 |
| 1 |  | 0 |

This truth table represents a NOT gate, where the output is the logical complement of the input. If the input is 0, the output will be 1, and if the input is 1, the output will be 0. Remember, a NOT gate, or inverter, produces the logical complement of its input. It flips the input value, producing the opposite value as the output.

31. The correct answer is b) 1. A NAND gate produces an output of 1 only when both inputs are 0. In this case, both inputs are 0, so the output of the NAND gate will be 1.

32. The correct answer is b) 2. A NAND gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

33. The correct answer is a) 0. A NAND gate produces an output of 0 when one input is 1 and the other input is 0. In this case, one input is 1

while the other is 0, so the output of the NAND gate will be 0.

34. The correct answer is d)  $(A * B)'$ . The Boolean expression  $(A * B)'$  represents the operation of a NAND gate, where  $*$  denotes the logical AND operation and  $'$  represents the logical complement or inversion.

35. The correct answer is a) 0. A NAND gate produces an output of 0 only when both inputs are 1. In this case, both inputs are 1, so the output of the NAND gate will be 0.

36. The correct answer is d) All of the above. In a truth table for a NAND gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 1. The output will be 1 for all of these combinations, except when both inputs are 1.

37. The correct answer is b) 1. If one or both inputs of a NAND gate are X (undefined), the output will be 1. This is because the output of a NAND gate is determined by the inputs, and if the inputs are undefined, the output is typically considered to be 1.

38. The correct answer is b) Rectangle. In logic circuit diagrams, a NAND gate is typically represented

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by a rectangle shape with two input lines and one output line.

39. The correct answer is a) The original input. Applying the NAND operation twice (NAND(NAND)) to an input will result in the original input value. The NAND gate inverts the inputs and produces an output, so applying it twice restores the original value.

40. The correct answer is c) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

This truth table represents a NAND gate, where the output is 1 for all combinations of inputs except when both inputs are 1. In that case, the output is 0.

Remember, a NAND gate is a combination of an AND gate followed by a NOT gate. It produces an output of 0 only when both inputs are 1; otherwise, it produces an output of 1.

41. The correct answer is b) 1. A NOR gate produces an output of 1 only when both inputs are 0. In this case, both inputs are 0, so the output of the NOR gate will be 1.

42. The correct answer is b) 2. A NOR gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

43. The correct answer is a) 0. A NOR gate produces an output of 0 when one input is 1 and the other input is 0. In this case, one input is 1 while the other is 0, so the output of the NOR gate will be 0.

44. The correct answer is d)  $(A + B)'$ . The Boolean expression  $(A + B)'$  represents the operation of a NOR gate, where + denotes the logical OR operation and ' represents the logical complement or inversion.

45. The correct answer is a) 0. A NOR gate produces an output of 0 only when both inputs are 1. In this case, both inputs are 1, so the output of the NOR gate will be 0.

46. The correct answer is d) All of the above. In a truth table for a NOR gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 0. The output will be 0 for all of these combinations, except when both inputs are 0.

47. The correct answer is b) 1. If one or both inputs of a NOR gate are X

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(undefined), the output will be 1.

This is because the output of a NOR gate is determined by the inputs, and if the inputs are undefined, the output is typically considered to be 1.

48. The correct answer is b)

Rectangle. In logic circuit diagrams, a NOR gate is typically represented by a rectangle shape with two input lines and one output line.

49. The correct answer is a) The original input. Applying the NOR operation twice ( $\text{NOR}(\text{NOR})$ ) to an input will result in the original input value. The NOR gate inverts the inputs and produces an output, so applying it twice restores the original value.

50. The correct answer is c)  $A \mid B$  |  
Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

This truth table represents a NOR gate, where the output is 1 for all combinations of inputs except when both inputs are 0. In that case, the output is 0.

Remember, a NOR gate produces an output of 1 only when both inputs are 0 otherwise, it produces an

output of 0. It is essentially the logical complement of the OR gate.

51. The correct answer is b) 1. An XOR gate produces an output of 1 when the inputs are different. In this case, both inputs are 0, so the output of the XOR gate will be 0.

52. The correct answer is b) 2. An XOR gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

53. The correct answer is b) 1. An XOR gate produces an output of 1 when the inputs are different. In this case, one input is 1 while the other is 0, so the output of the XOR gate will be 1.

54. The correct answer is c) A XOR B. The Boolean expression  $A \text{ XOR } B$  represents the operation of an XOR gate. XOR stands for exclusive OR, indicating that the output is 1 only when the inputs are different.

55. The correct answer is b) 1. An XOR gate produces an output of 1 when the inputs are different. In this case, both inputs are 1, so the output of the XOR gate will be 0.

56. The correct answer is d) All of the above. In a truth table for an

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XOR gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 1. The output will be 0 for the first and last combinations and 1 for the second combination.

57. The correct answer is b) 1. If one or both inputs of an XOR gate are X (undefined), the output will be 1. This is because the output of an XOR gate is determined by the inputs, and if the inputs are undefined, the output is typically considered to be 1.

58. The correct answer is d) Diamond. In logic circuit diagrams, an XOR gate is typically represented by a diamond shape with two input lines and one output line.

59. The correct answer is a) The original input. Applying the XOR operation twice ( $XOR(XOR)$ ) to an input will result in the original input value. The XOR gate compares the inputs and produces an output, so applying it twice restores the original value.

60. The correct answer is b) A | B | Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

This truth table represents an XOR gate, where the output is 1 only when the inputs are different. In all other cases, the output is 0.

Remember, an XOR gate produces an output of 1 only when the inputs differ; otherwise, it produces an output of 0. It is essentially the logical exclusive OR operation.

61. The correct answer is b) 1. An XNOR gate produces an output of 1 when both inputs are the same. In this case, both inputs are 0, so the output of the XNOR gate will be 1.

62. The correct answer is b) 2. An XNOR gate typically has two inputs. It requires two signals or binary values as input and produces a single output.

63. The correct answer is a) 0. An XNOR gate produces an output of 0 when the inputs are different. In this case, one input is 1 while the other is 0, so the output of the XNOR gate will be 0.

64. The correct answer is c) A XNOR B. The Boolean expression  $A \text{ XNOR } B$  represents the operation of an XNOR gate. XNOR stands for exclusive NOR, indicating that the output is 1 when the inputs are the same.

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65. The correct answer is b) 1. An XNOR gate produces an output of 1 when both inputs are the same. In this case, both inputs are 1, so the output of the XNOR gate will be 1.

66. The correct answer is d) All of the above. In a truth table for an XNOR gate, there are three possible combinations of inputs: 0 and 0, 0 and 1, and 1 and 1. The output will be 1 for the first and last combinations and 0 for the second combination.

67. The correct answer is b) 1. If one or both inputs of an XNOR gate are X (undefined), the output will be 1. This is because the output of an XNOR gate is determined by the inputs, and if the inputs are undefined, the output is typically considered to be 1.

68. The correct answer is d) Diamond. In logic circuit diagrams, an XNOR gate is typically represented by a diamond shape with two input lines and one output line.

69. The correct answer is a) The original input. Applying the XNOR operation twice ( $\text{XNOR}(\text{XNOR})$ ) to an input will result in the original

input value. The XNOR gate compares the inputs and produces an output, so applying it twice restores the original value.

70. The correct answer is b) A | B |  
Output

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

This truth table represents an XNOR gate, where the output is 1 only when the inputs are the same. In all other cases, the output is 0.

Remember, an XNOR gate produces an output of 1 when the inputs are the same; otherwise, it produces an output of 0. It is essentially the logical exclusive NOR operation.